

# Ernest Ting

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*Applied Computational Mathematics and Business, Economics, & Management @ Caltech*

## EDUCATION

**California Institute of Technology (Caltech)**, Pasadena, CA | **2025 - 2029**

- **Activities:** NCAA DIII Cross-Country & Track | Peer Advocate | Peer Academic Coach | Chamber Music | Health Advocate (In Training)
- **Relevant Coursework:** Computer Programming, Analytical Calculus & Linear Algebra, Mechanics & Electromagnetism, Microeconomics and Macroeconomics, Economic Methodology, Biological Processes, Psychology, Neuroscience

**Brookings High School**, Brookings, SD | **2021 - 2025**

- **Activities:** Captain, Science Olympiad | Math Club | Debate | Captain, Quiz Bowl | Concertmaster, Orchestra | 4x Team Captain, Varsity Cross-Country, Swimming & Track | Peer Tutor (Math)
- **Relevant Coursework:** Multivariable Calculus/Linear Algebra, Vector Calculus/Differential Equations, Data Analysis, Biomedical Engineering, AP Physics, AP Calculus, AP Biology, AP Statistics, AP Microeconomics, AP Macroeconomics

### *Honors & Awards*

U.S. Presidential Scholar | U.S. Naval Horizons Highest Honor | South Dakota Science Olympiad Champion | South Dakota Mathcounts Champion | 4x All-State Orchestra | 4x800 State Champion Track | 3x State Champion Swimming | 17x Varsity Letters

## RESEARCH & WORK EXPERIENCE

**Summer Researcher, Caltech, INNODEMS - 2025:** Formulated and investigated methods using classical item response analysis to evaluate student performance, item difficulty, and potential response tree (PRT) failure modes across STACK online math assessments.

**Waiter, Caltech Dining Services - 2026-Present:** Assisted with setup, distribution of food, and cleanup of House Dinners.

**Research Assistant, South Dakota State University - 2025:** Assisted graduate students in research regarding combinatorics, specifically exploring Ramsey's Theorem, Schur's Theorem, and monochromatic subgraphs within edge-colored graphs. Conducted literature reviews of recent combinatorial proofs and strategies, and compiled data into weekly briefings for graduate students.

**Research Assistant, South Dakota State University - 2024:** Assisted with bioinformatics research, specifically gene set analysis, as part of a larger NIH-funded project. Used computational methods to analyze differences in gene transcript length and find differences between genes and the gene sets that include them. Found a correlation between gene transcript length and certain cancers.

**Intern, YMCA Camp Widjiwagan - 2025-2026:** Led multi-day wilderness canoe trips for youth in the Boundary Waters Canoe Area Wilderness. Nurtured the personal growth and confidence of campers through hands-on learning and nature immersion. Managed all trip logistics, including route navigation, equipment maintenance, and risk management protocols in remote wilderness environments.

**Swim Instructor, Brookings Swim Club - 2021-2025:** Taught swim fundamentals and water safety to elementary-aged swimmers. Built confidence and technique through clear communication and positive reinforcement. Evaluated student performance against Red Cross and Swim America national certification standards and provided progress reports to parents to ensure goal alignment.

### *Volunteer Work*

**Volunteer, Union Station Homeless Center - 2025-Present:** Assist with meal preparation by washing and chopping vegetables, preparing ingredients, and supporting kitchen operations to help provide meals for community members.

**Distributor, Feeding America - 2024-2025:** Helped facilitate the weekly distribution of 400 food boxes to over 1,000 community members in need with a team of volunteers. Streamlined the assembly and loading process and assisted in monitoring inventory levels.

**Math Tutor, MATHCOUNTS - 2021-2025:** Provided individual and small-group tutoring to middle school students. Helped to strengthen their foundational math skills, competition problem-solving, and preparation for MATHCOUNTS competitions.

## PROJECTS

- Developing a Moodle/STACK plugin enabling embedded, real-time analytics within course modules for instructors.
- Training multi-agent LLMs to generate STACK assessment questions and textbook content for Caltech's Math 1b/1c curriculum.
- Developed an interactive Streamlit and Plotly web application to extract statistics, item difficulty metrics, and potential response tree pass rates from Moodle and STACK online mathematics assessment data.
- Created a Discord Bot that took the daily dining hall food special and messaged in the house Discord.
- Designing and developing a social media app that allows outdoor enthusiasts to share catches and hunts, gear reviews, photos, and popular spots.

## PRESENTATIONS

- **Poster Presentation**, International STACK Conference, Nairobi, Kenya: "Turning STACK Response Data into Insight, July 2026"
- **Slideshow Presentation**, International STACK Conference, Nairobi, Kenya: "Moodle/STACK Interactive Quiz Analytics", July 2026
- **Oral Presentation**, Original Research, 2025 BHS Literary Conference: "The Big Lie: Distilling the Truth from Power", April 2025

## TECHNICAL SKILLS

**Programming & Other:** Python | R | Java | LaTeX | ggplot | ggplot2 | MATLAB | Microsoft Office Suite | Google Workspace | Streamlit | Pandas | Plotly | Kaleido | SciPy | Matplotlib | mathtext | ReportLab | OpenPyXL | Playwright | Chromium |